

Physics-Informed Machine Learning for Robust Building Energy Prediction under Incomplete Sensor Data

MSc Research Project

Sathish Kumar Konda

Student ID: 25190890

MSc in Artificial Intelligence
National College of Ireland

September 2026

Abstract

Building energy models are trained on sensor streams that, in operation, are often incomplete: sensors fail, controllers drop off the network and some points are never installed. This project tests whether a physics-informed machine-learning model predicts hourly HVAC energy more reliably than conventional models when input sensors are missing. The study uses the public three-year dataset of LBNL Building 59, an office building in Berkeley, California, and adopts the target, test window (July–December 2020) and metrics of the principal baseline, the Bayesian neural network of Mahajan et al. (2024), which assumes complete, forward-filled data.

The proposed PI-GRU is a mask-aware recurrent network that jointly predicts HVAC energy and indoor temperature and is regularised by a first-order resistance–capacitance heat balance with learnable, positive coefficients. It is compared with a re-implementation of the baseline BNN, a sensor-free profile, XGBoost, LSTM and GRU models and two ablations, under identical random (MCAR) and contiguous multi-sensor outages that remove 10–40% of the readings, with five seeds, day-block bootstrap intervals and Holm-adjusted Wilcoxon tests.

On complete data the PI-GRU beats the published baseline on every metric: RMSE 9.01 against 9.65 kWh, MAE 5.70 against 7.06 kWh and MAPE 0.22 against 0.35. Under 40% contiguous outages it is 20% more accurate than the re-implemented baseline. XGBoost on the same mask-aware features has the lowest RMSE (7.97 kWh), and the physics term is neutral for accuracy relative to an architecture-matched twin, while lowering the heat-balance residual by 10% on complete data. Random sensor loss barely affects any model, contiguous outages raise errors by 15–23%, and observation-mask features are the most effective single protection. A least-squares fit shows that the envelope time constant is not identifiable from this tightly controlled building, which explains the weak contribution of the physics prior. The code, data and a notebook that runs on Colab, Jupyter, macOS and Windows are released.

Keywords: physics-informed machine learning, HVAC energy prediction, missing sensor data, smart buildings, robustness, LBNL Building 59

Contents

1	Introduction	4
1.1	Background	4
1.2	Problem statement and baseline	4
1.3	Research question and objectives	5
1.4	Contributions	5
1.5	Structure of the report	5
2	Related Work	5
2.1	Data-driven building energy prediction	5
2.2	Missing data in building datasets	6
2.3	Physics-informed models of building thermal dynamics	6
2.4	Physics-informed imputation and energy prediction	7
2.5	Critical evidence map and research gap	7
3	Research Methodology	8
3.1	Research design	8
3.2	Dataset	8
3.3	Tasks and split	9
3.4	Controlled sensor loss	10
3.5	Evaluation	10
3.6	Fair comparison	11
3.7	Ethical considerations	11
4	Design Specification	11
4.1	Overview	11
4.2	Physics prior	12
4.3	Proposed model: PI-GRU	12
4.4	Baselines and ablations	13
4.5	Alternatives considered	14
5	Implementation	14
6	Evaluation	15
6.1	Comparison with the baseline paper	15
6.2	Robustness to sensor loss	17
6.3	Is the gain due to physics? Statistical tests and ablations	18
6.4	Physical consistency and learned parameters	19
6.5	Operating strata	20
6.6	Choice of imputer	21
6.7	Computational cost	22

6.8	Supplementary: forecasting with meter history	22
6.9	Summary against the objectives and hypotheses	23
7	Discussion	23
7.1	Answer to the research question	23
7.2	Why the physics prior did not help here	24
7.3	Comparison with the literature	24
7.4	Implications	24
7.5	Limitations	25
8	Conclusion and Future Work	25

1 Introduction

1.1 Background

Office buildings now carry dense sensor networks. Building-management systems log electricity use, supply- and return-air temperatures, fan speeds, indoor temperatures and on-site weather every few minutes. Facility managers use these streams to predict heating, ventilation and air-conditioning (HVAC) energy, to detect faults and to plan demand response. Buildings account for a large share of electricity use, and HVAC is usually the largest controllable end use, so accurate HVAC prediction has direct cost and carbon value (Amasyali and El-Gohary, 2018).

Real sensor data are rarely complete. Sensors fail, controllers drop off the network, loggers drift, and some points are never installed because they are expensive or intrusive. The public three-year dataset of LBNL Building 59 (Luo et al., 2022), used in this project, shows the problem clearly. Even after the authors’ curation, the supply-air temperature is missing for 7.9% of the hours, the HVAC meter for 4.6% and the Wi-Fi occupancy counts for more than half of the period (Section 3.2).

Purely data-driven models learn statistical associations between inputs and energy. When an input disappears at run time, the model sees a pattern it was never trained on, and its prediction can become both inaccurate and physically implausible. Physics-informed machine learning (PIML) adds prior physical knowledge, such as an energy balance, to the architecture or to the loss function (Karniadakis et al., 2021; Raissi et al., 2019). For buildings, a first-order resistance–capacitance (RC) model links indoor temperature change to outdoor temperature, internal gains, solar gains and HVAC power (Bacher and Madsen, 2011). Such a prior can constrain a neural network when parts of the data are missing.

1.2 Problem statement and baseline

The principal baseline of this project is Mahajan et al. (2024). That study predicts hourly HVAC energy of LBNL Building 59 from weather, indoor temperature, other electricity loads and calendar variables with a Bayesian neural network (BNN). It trains on about 2.5 years and tests on the last six months of 2020, and reports a test RMSE of 9.65 kWh, an MAE of 7.06 kWh and a MAPE of 0.35 for the BNN (11.86, 8.21 and 0.45 for an MC-dropout LSTM). This is the closest published work: same building, same target, same test window.

The baseline has three gaps that this project addresses:

1. **Complete data only.** Missing values are forward-filled before modelling and the models are never tested with missing sensors. Its robustness to sensor loss is unknown.
2. **No physical knowledge.** The BNN is a generic multilayer perceptron. Nothing ties its predictions to the heat balance of the building.
3. **No temporal context and no missingness signal.** Each hour is predicted from that hour’s readings only, and an imputed value looks the same to the model as a measured one.

Physics-informed building models exist (Gokhale et al., 2022; Di Natale et al., 2022; Chen et al., 2023), and so do studies of missing building data (Kim et al., 2023; Lee et al., 2024; Liguori et al., 2024). They are, however, rarely brought together: no study found in this review tests whether a physics prior protects an operational HVAC energy predictor when its input sensors fail (Section 2.5).

1.3 Research question and objectives

Research question. *To what extent does a mask-aware, physics-informed recurrent network improve hourly HVAC energy prediction over conventional machine-learning models, including the published baseline, when the building’s input sensors are incomplete?*

Sub-questions.

RQ1 How does accuracy change as 10–40% of the sensor readings are removed?

RQ2 Does the answer differ between random point loss (MCAR) and contiguous multi-sensor outages?

RQ3 Does the physics term reduce physical inconsistency (balance residual, implausible responses), and are the learned thermal parameters credible?

Objectives.

- O1 Build a reproducible benchmark with a fixed chronological split and identical, seeded missingness masks for every model.
- O2 Re-implement the baseline BNN and implement strong conventional baselines (profile, XGBoost, LSTM, GRU) and the proposed physics-informed GRU (PI-GRU) with architecture-matched ablations.
- O3 Evaluate accuracy, robustness, physical consistency and computational cost with five seeds, bootstrap confidence intervals and paired tests.
- O4 Beat the published baseline on its own protocol and state when the physics prior helps, is neutral, or hurts.

1.4 Contributions

- A sensor-loss benchmark for Building 59 with MCAR and contiguous outages at four rates, applied identically to every model, plus a meter-outage stress test.
- A mask-aware PI-GRU that jointly predicts HVAC energy and indoor temperature and is regularised by a learnable RC heat balance with positive coefficients.
- A faithful re-implementation of the baseline BNN, whose complete-data errors fall close to the published ones, and a direct comparison on the baseline’s protocol.
- Evidence, with confidence intervals and ablations, on where the accuracy gain comes from: the temporal, mask-aware architecture, the auxiliary temperature task or the physics term.
- An open, one-command pipeline and a notebook that runs on Colab, Jupyter, macOS and Windows with CUDA, Apple-silicon MPS or CPU.

1.5 Structure of the report

Section 2 reviews related work and identifies the gap. Section 3 describes the research design, data and evaluation protocol. Section 4 specifies the models, and Section 5 the implementation. Section 6 presents the results and Section 7 discusses them. Section 8 concludes.

2 Related Work

2.1 Data-driven building energy prediction

Data-driven prediction of building energy use is a mature field. Amasyali and El-Gohary (2018) review more than a hundred studies and find that artificial neural networks and support vector regression dominate, that most studies predict at hourly or daily resolution, and that few test models outside the conditions they were trained on. Gradient-boosted trees (Chen and Guestrin,

2016) and recurrent networks (Hochreiter and Schmidhuber, 1997; Cho et al., 2014) have since become standard strong baselines for tabular and sequential energy data.

Luo et al. (2022) published the dataset used here: three years (2018–2020) of energy, HVAC, indoor-environment, weather and occupancy data from LBNL Building 59, an office building in Berkeley, California, with more than 300 points. They document the raw gaps and release a cleaned version, which shows that missing data are a property of the system, not an artefact.

Principal baseline. Mahajan et al. (2024) use this dataset to predict hourly HVAC and total electricity demand. Their inputs are season, weekend and duty-hour flags, a maintenance-event flag, average interior zone temperature, solar radiation, relative humidity, air temperature, wind speed and electricity load. Missing values are filled by copying the previous hour. They train a Bayesian neural network (three layers of 512, 512 and 128 units, trained by variational inference (Blundell et al., 2015)) and an MC-dropout LSTM (Gal and Ghahramani, 2016) on about 2.5 years and test on the last six months of 2020, with a quantile random forest as benchmark. For HVAC on the full test set they report RMSE 9.65 kWh, MAE 7.06 kWh and MAPE 0.35 (BNN), 11.86, 8.21 and 0.45 (MC-LSTM) and 21.67, 17.17 and 0.40 (QRF). Their focus is uncertainty quantification. The limits that matter here are that the evaluation assumes complete, forward-filled data, the model contains no physical knowledge and each hour is predicted without temporal context. This study adopts their target, split and metrics so that results are directly comparable, and then asks what happens when the sensors fail.

2.2 Missing data in building datasets

RUBIN (1976) distinguishes data missing completely at random (MCAR) from data missing at random or not at random. In buildings, loss is rarely MCAR: a controller outage removes several points for hours or days. Kim et al. (2023) compare imputation methods for residential energy, behaviour and environment data and find that the best method depends on gap length and variable type. Lee et al. (2024) compare median, k -nearest-neighbour, CART and XGBoost imputation for building energy benchmarking and show that the imputation choice changes both the reconstructed distribution and the downstream model. Both studies judge imputation by reconstruction or benchmark quality, not by the robustness of a downstream predictor.

In machine learning more broadly, Che et al. (2018) show that recurrent networks can exploit missingness itself: feeding the observation mask and the time since the last observation (GRU-D) improves prediction on clinical time series. This project borrows that idea: every model receives the mask and a time-since-last-observation feature, so it can tell a measured value from a carried-forward one.

2.3 Physics-informed models of building thermal dynamics

Physics-informed learning combines data with physical laws through soft penalties, constrained architectures or hybrid models (Raissi et al., 2019; Karniadakis et al., 2021). Karniadakis et al. (2021) note that physics adds information beyond the labels but that balancing data and physics losses of different scales is difficult. Grey-box RC models are the standard low-order physics of buildings; Bacher and Madsen (2011) show how to select such models for the heat dynamics of a building from data.

Gokhale et al. (2022) encode thermal dynamics in a PINN that predicts room temperature, energy and an unobserved thermal-mass state; it needs less training data and extrapolates further than a plain network. Di Natale et al. (2022) propose Physically Consistent Neural Networks whose structure guarantees a sensible response to heating inputs and to neighbouring and outdoor temperatures, with accuracy similar to black-box networks. Chen et al. (2023) integrate a thermal model with a neural network (PhysCon) for thermal modelling and demand-response

control. Structural constraints give stronger guarantees but need zone-level power and building parameters that public operational data rarely provide. Ma et al. (2025) review PIML for building energy and name the balance between accuracy and physical consistency, the choice of physics prior and validation on real buildings as open problems. None of these studies removes sensors at inference time.

2.4 Physics-informed imputation and energy prediction

Liguori et al. (2024) couple a denoising autoencoder with a thermal-balance equation to impute temperature, heating and cooling data. The physics term barely changes reconstruction error on complete data but improves robustness as the missing rate grows and yields interpretable coefficients. This suggests that the benefit of physics lies in graceful degradation, not in best-case accuracy. Their endpoint is imputation, not energy prediction. Michalakopoulos et al. (2024) use a PINN whose network estimates envelope and thermal properties that feed heat-loss equations, predicting the energy efficiency of 256 residential buildings from static audit data. It is an energy-prediction PINN, but with static inputs and no sensor loss.

2.5 Critical evidence map and research gap

Table 1 summarises the evidence. Each stream supplies a piece of the design; none combines operational HVAC prediction, physics and controlled sensor loss.

Table 1: Critical evidence map.

Study	Focus	Used in this project	Limitation creating the gap
Mahajan et al. (2024)	BNN / MC-LSTM, HVAC of Bldg 59	Target, split, metrics; BNN re-implemented as principal baseline	Complete forward-filled data; no physics; no temporal context
Luo et al. (2022)	Bldg 59 dataset	Data source	Curated data need controlled re-masking
Kim et al. (2023); Lee et al. (2024)	Missing-data handling	Imputers (LOCF, mean, CART)	No downstream physics-informed comparison
Che et al. (2018)	RNNs with missing values	Mask and time-since-last features	Not physics-aware; clinical data
Gokhale et al. (2022); Di Natale et al. (2022); Chen et al. (2023)	Physics-guided thermal models	RC balance, positive coefficients	Sensor-outage robustness not evaluated
Liguori et al. (2024)	Physics-informed imputation	Soft thermal residual; robustness framing	Imputation, not energy prediction
Michalakopoulos et al. (2024)	PINN energy prediction	PINN energy framing	Static audit inputs; no sensor loss
Ma et al. (2025)	PIML review	Open problems	Calls for real-building validation

Gap. There is no controlled test of whether an approximate building-physics prior reduces the error of an operational HVAC energy predictor when its input sensors fail, and no such test against the published baseline for the same building. This project fills that gap with identical masks for every model, random and contiguous outages, an architecture-matched ablation that isolates the physics term, and joint reporting of accuracy, robustness slope and physical consistency.

3 Research Methodology

3.1 Research design

The study is a quantitative computational experiment in the design-science tradition: an artefact (the PI-GRU) is built and compared with known alternatives under controlled conditions. A case study would describe one deployment without controlled comparison, and a purely analytical study could not show whether an approximate physics prior helps on noisy operational data. The independent variables are the model, the missingness pattern (MCAR or contiguous block) and the missingness rate (0, 10, 20, 30, 40%). The dependent variables are MAE, RMSE, MAPE, R^2 , NRMSE, the RMSE increase over complete data and its slope, the RC-balance residual, the share of physically implausible responses, and training and inference time.

Hypotheses. H1: at 20–40% missing readings the PI-GRU has a lower RMSE than the best conventional model. H2: the PI-GRU has a lower RMSE than its architecture-matched twin without the physics term ($\lambda = 0$). H3: the physics term lowers the RC-balance residual without raising complete-data RMSE by more than 2%. The success criterion fixed before the experiments was a reduction of at least 5% in RMSE against the best conventional model at one or more rates between 20% and 40%. It is used to interpret results, not to hide contrary ones.

3.2 Dataset

The data are the cleaned release of LBNL Building 59 (Luo et al., 2022; Hong et al., 2022), obtained through a Kaggle mirror of the Dryad files because Dryad blocks scripted downloads. The files used are the whole-building electricity meters, the site weather station, the 16 indoor temperature loggers and the four rooftop units’ supply-air temperature, return-air temperature and supply-fan speed. All series are averaged to hourly values on a UTC index; calendar variables use local time (America/Los_Angeles). Table 2 lists the resulting variables.

Table 2: Hourly variables used in this study.

column	unit	source	description
hvac_kwh	kWh	ele.csv hvac_N + hvac_S	HVAC electricity in the hour (forecast target, also lagged input)
t_in	°C	zone_temp_interior.csv	Median of 16 indoor air-temperature loggers
t_out	°C	site_weather.csv	Outdoor air temperature, LBNL weather station
rh_out	%	site_weather.csv	Outdoor relative humidity
solar	W/m ²	site_weather.csv	Global solar radiation
q_int	kW	solar_radiation_set_1 ele.csv mels_N + mels_S + lig_S	Plug-load and lighting power, internal heat-gain proxy
t_sa	°C	rtu_sa_t.csv mean of 4 RTUs	Supply-air temperature
t_ra	°C	rtu_ra_t.csv mean of 4 RTUs	Return-air temperature
fan	%	rtu_fan_spd.csv supply-fan VFD mean	Supply-fan speed feedback
local_hour	h	derived	Hour of day, America/Los_Angeles
local_dow	0-6	derived	Day of week (Monday = 0), local time

Three design decisions follow from the data. First, the Wi-Fi and camera occupancy counts are missing for most of August 2018–February 2020 in the release, so plug-load plus lighting power is

used as the internal-gain proxy; it is also the physically direct quantity, because electricity used indoors ends up as heat. Second, supply-air readings of 0 °F and fan speeds outside 0–100% are treated as missing. Third, gaps of up to three hours are filled by time interpolation inside the curated table; longer natural gaps stay missing and are never used as targets. After this, 86% of the hours are complete; natural gaps are mostly in the supply-air temperature in 2018 and in the HVAC meter (Table 3).

Table 3: Natural gaps and correlation with HVAC energy, 2018-03-01 to 2020-12-31 (24,888 hours).

channel	natural gaps (%)	after short-gap filling (%)	correlation with HVAC
hvac_kwh	5.02	4.85	1.00
t_in	0.46	0.42	0.22
t_out	0.00	0.00	0.45
rh_out	0.00	0.00	-0.23
solar	0.00	0.00	0.24
q_int	3.51	3.36	0.25
t_sa	9.78	9.41	0.11
t_ra	0.00	0.00	0.30
fan	0.82	0.74	0.51

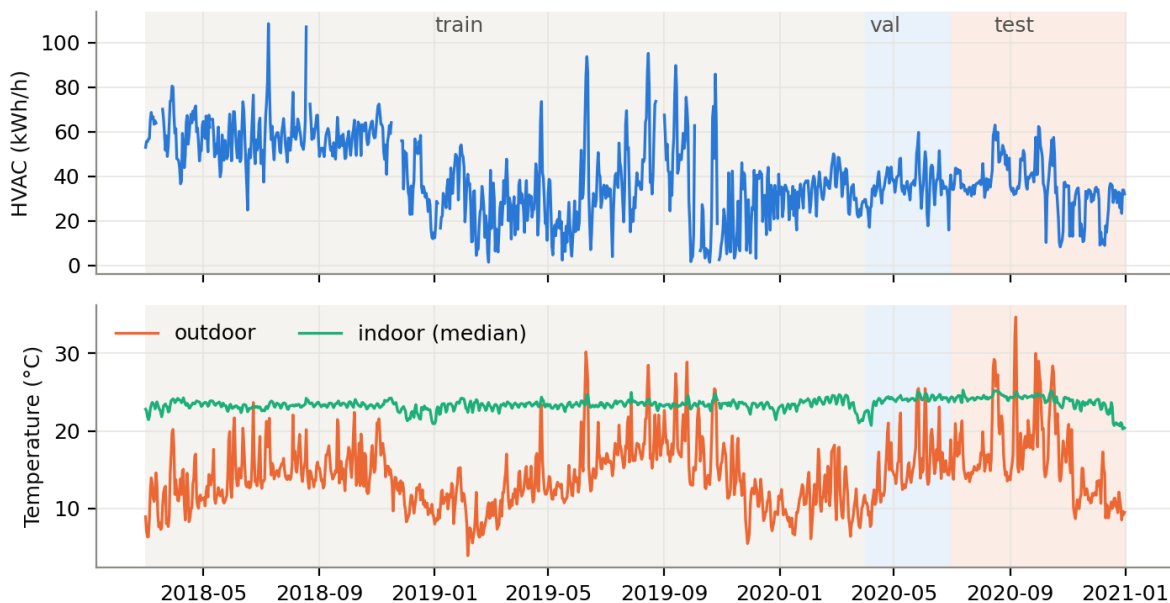


Figure 1: Daily mean HVAC energy (top) and outdoor and indoor temperature (bottom), with the chronological split.

3.3 Tasks and split

Estimation (primary task). Following Mahajan et al. (2024), the model predicts the HVAC energy of hour t from the building and weather sensors up to and including hour t , without the HVAC meter history. This is the setting in which sensor loss matters: the model has to infer HVAC energy from the building’s thermal state and drivers, which is what a physics prior describes. It is also the setting of a virtual meter or of a model that runs on forecast weather.

Forecasting (supplementary task). The model predicts the next hour’s HVAC energy with the meter history as an extra input. This shows how much the meter history dominates and adds a stress test in which the meter itself is lost.

Split. The split reproduces the baseline’s test window: training 1 March 2018–31 March 2020, validation 1 April–30 June 2020, test 1 July–31 December 2020 (4,414 hourly targets). Training starts in March 2018 because the indoor loggers start on 22 February 2018. Model selection uses the validation quarter only. Every model is then refitted on training plus validation for the selected number of epochs or trees, so the final models, like the baseline’s, see about 2.5 years of data; the test set is used once. All scaling and imputation statistics come from the training rows only.

3.4 Controlled sensor loss

Masks are applied to the eight sensor channels after the split, on top of the natural gaps. The HVAC meter is masked only in the forecasting stress test.

- **MCAR:** each observed reading is removed independently with probability r .
- **Block outages:** outages of 6–72 hours, each hitting one to three channels at once (for example a rooftop-unit controller losing supply-air temperature, return-air temperature and fan speed together), are drawn until a share r of the observed readings is removed.

Rates are 10, 20, 30 and 40%, with five seeds. The mask for a given seed, pattern and rate is the same for every model, so all comparisons are paired. Figure 2 shows the two patterns.

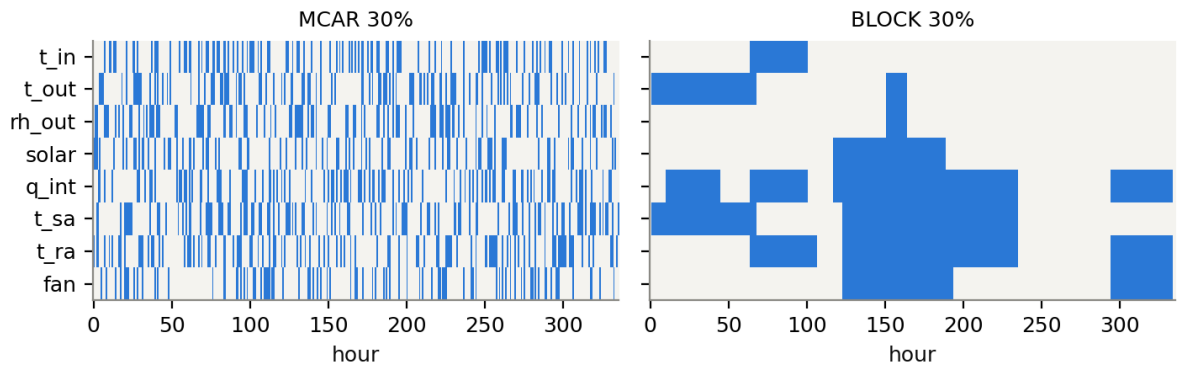


Figure 2: Two weeks of the MCAR and block masks at 30% (blue = removed).

Missing readings are imputed causally by last observation carried forward (LOCF), the method the baseline uses, with the training mean before the first observation. Mean and iterative CART imputation are tested as alternatives. Each channel also gets a binary observation mask and the scaled time since its last observation (Che et al., 2018). Models are trained on a pool of nine versions of the training data (complete, and MCAR and block masks at each rate, with training-only seeds), so that every model learns under the same missingness it is later tested on.

3.5 Evaluation

Accuracy is measured by MAE, RMSE, MAPE (hours with at least 1 kWh, to avoid division by near-zero), R^2 and NRMSE (RMSE over mean load). Robustness is the RMSE increase over complete data and the least-squares slope of RMSE against the missing rate, in kWh per 10 percentage points. Physical consistency is measured three ways: the RMS residual of a least-squares RC balance fitted on clean training data, applied to the dual-head models’ predicted temperature and energy; the change in predicted HVAC energy when the outdoor temperature input is raised by 1 K; and the share of cooling hours in which that change is negative, which is physically implausible because warmer outdoor air raises the cooling load. Training and inference time record the cost of the physics.

Every model is trained with five seeds. For each condition, the PI-GRU is compared with the best conventional model at that condition and with its $\lambda = 0$ twin. The 95% confidence interval of the RMSE difference is a day-block bootstrap (Kunsch, 1989) with 2,000 resamples on seed-averaged squared errors, which respects the dependence between hours of the same day. A Wilcoxon signed-rank test on daily MSE gives a p -value, and Holm’s method (Holm, 1979) adjusts for the number of conditions. Errors are also split by occupied and unoccupied hours and by mild and extreme outdoor temperature.

3.6 Fair comparison

All trainable models use the same data, features, masks, split and early-stopping rule (patience 6 on mean validation RMSE over five validation conditions). Each model family gets the same tuning budget of five validation trials: XGBoost over depth and learning rate, the BNN over the baseline’s own learning-rate and batch-size grid, the LSTM and the GRU over hidden size, learning rate, weight decay and dropout. The GRU architecture is tuned on the $\lambda = 0$ dual-head model, so the physics term has no influence on it; the PI-GRU then tunes only λ over five values. The ablations reuse the PI-GRU’s configuration.

3.7 Ethical considerations

No people were recruited and no personal data were collected. The dataset is public and licensed for reuse (CC0 on Dryad), and it is cited as required. The Wi-Fi and camera counts, the only person-related data in the release, are not used; the internal-gain proxy is electricity. The models are presented as decision-support tools, not as autonomous HVAC controllers, and their limits are reported. Experiments were kept small (early stopping, five-trial tuning, laptop hardware) to limit energy use. Generative-AI assistance used in preparing code and text is declared in the submission as required by NCI policy.

4 Design Specification

4.1 Overview

Figure 3 shows the pipeline. The hourly table passes through the mask generator, the causal imputer and the feature builder, which emits, for every hour, the eight standardised sensor values, their eight observation masks, their eight scaled times since last observation and five calendar terms (hour and weekday as sine/cosine pairs and a weekend flag). All models consume these same features; the recurrent models read a 24-hour window, the tabular models read the target hour plus the previous hour’s sensor values.

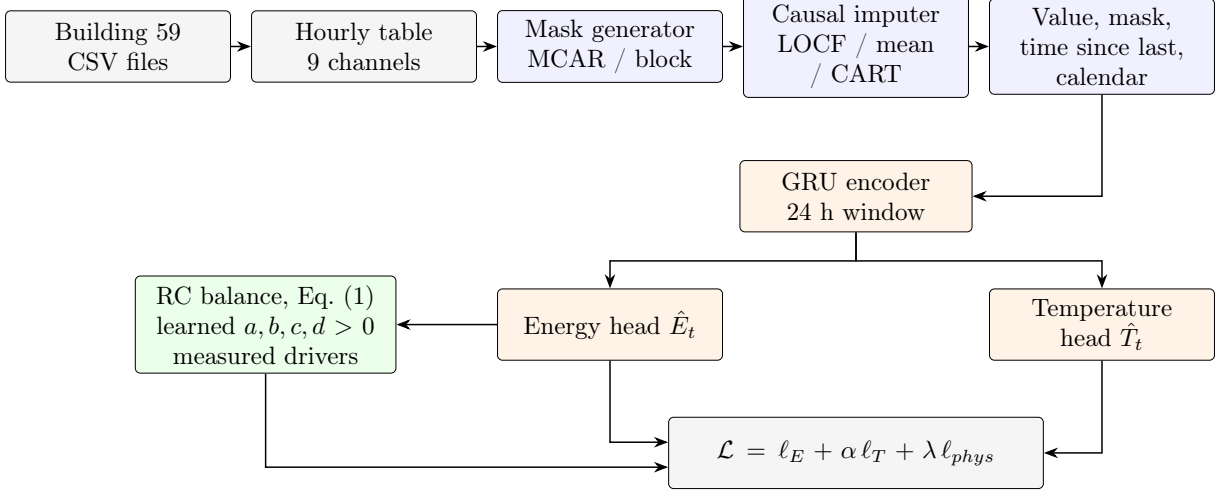


Figure 3: Pipeline and PI-GRU architecture. Blue: shared data path used by every model. Orange: network. Green: physics term, active only during training.

4.2 Physics prior

A first-order RC model treats the building as one thermal capacitance C connected to outdoor air through a conductance UA and heated or cooled by internal gains, solar gains and the HVAC system (Bacher and Madsen, 2011). With a one-hour step and hourly means,

$$T_t - T_{t-1} = a(T_t^{out} - T_{t-1}) + b m_t E_t + c Q_t^{int} + d S_t + e, \quad (1)$$

where T is the indoor temperature, E the HVAC energy, Q^{int} the plug-load and lighting power, S the solar radiation (kW/m^2), $a = UA/C$, $b = \eta/C$, $c = \beta/C$ and $d = \gamma/C$. The HVAC mode $m_t = \tanh((T_t^{sa} - T_t^{ra})/2)$ is positive when the rooftop units supply air warmer than the return air (heating) and negative when they cool; it is close to zero in fan-only operation. The coefficients a, b, c, d pass through a softplus, so they are always positive, and e absorbs unmodelled constant gains. The time constant $\tau = 1/a$ is reported as an interpretability check.

The balance is deliberately simple. Building 59 has four rooftop units, fan-coil units and a central chilled- and hot-water plant (Mahajan et al., 2024); only the rooftop-unit electricity is in the HVAC meter, and zone-level flows are not available. A hard structural constraint would therefore be mis-specified. A soft residual with a tuned weight lets the data override the prior where it is wrong (Karniadakis et al., 2021; Ma et al., 2025).

4.3 Proposed model: PI-GRU

The PI-GRU encodes the 24-hour feature window with a GRU (Cho et al., 2014), applies dropout and a ReLU layer, and has two linear heads: HVAC energy $\hat{E}_t$ and indoor temperature $\hat{T}_t$. The loss is

$$\mathcal{L} = \text{Huber}(\hat{E}_t, E_t) + \alpha \text{Huber}(\hat{T}_t, T_t) + \lambda \frac{\sum w_t \text{Huber}(r_t)}{\sum w_t}, \quad r_t = \frac{(\hat{T}_t - T_{t-1}) - \text{RHS}_t(\hat{E}_t)}{\sigma_{\Delta T}}, \quad (2)$$

with targets standardised on the training set, $\alpha = 1$, and RHS_t the right-hand side of Eq. (1) evaluated with the predicted energy. Two design choices make the term useful under sensor loss:

- **The residual uses measured drivers.** During training the true outdoor temperature, internal gains, solar radiation, mode and previous indoor temperature are known, even when the network’s own inputs are masked. The physics term therefore teaches the network

to output energy–temperature pairs that are consistent with the real thermal state, which is exactly the information it has to reconstruct when sensors fail. At test time the residual is not needed.

- **The residual is mask-aware.** The weight w_t is zero wherever a driver of Eq. (1) is naturally missing, so the prior is never enforced on fabricated values.

4.4 Baselines and ablations

- **Profile:** mean HVAC energy for the same hour of day and weekday in the training data; needs no sensors.
- **BNN-paper:** re-implementation of the baseline (Mahajan et al., 2024): a mean-field Bayesian network (Blundell et al., 2015) with layers of 512, 512 and 128 units, a Gaussian likelihood with softplus scale, the negative log-likelihood plus KL loss, log-transformed and standardised target, and the baseline’s inputs (indoor temperature, outdoor temperature, humidity, solar radiation, other electricity, season and calendar flags, duty hours). Wind speed and the maintenance-event flag are not in the released files. Predictions average 30 posterior samples.
- **BNN:** the same network on this project’s features (with masks and previous-hour values), to separate the feature set from the model class.
- **XGBoost:** gradient-boosted trees (Chen and Guestrin, 2016) on the tabular features.
- **LSTM** (Hochreiter and Schmidhuber, 1997) and **GRU:** energy-only recurrent networks on the same window.
- **GRU-aux:** the PI-GRU with $\lambda = 0$. It has the same architecture, inputs and auxiliary temperature head, so the difference between PI-GRU and GRU-aux isolates the physics term.
- **PI-GRU-nomask:** the PI-GRU without mask and time-since-last features, which isolates mask awareness.

For the forecasting task, persistence (last hour’s energy) and seasonal persistence (same hour yesterday) replace the profile, and all models also receive the meter channel.

4.5 Alternatives considered

Table 4: Design alternatives and the reason for the choice made.

Decision	Alternatives	Choice and reason
Physics integration	Hard structure (PCNN, PhysCon), hybrid simulation	Soft residual: zone flows and plant power are unobserved, so a hard structure would be mis-specified; λ can be tuned and ablated.
Thermal model order	2R2C or higher-order RC	1R1C: identifiable from building-level data; higher orders need wall or zone temperatures that are not in the release.
Backbone	Temporal MLP, transformer	GRU: fewer parameters than an LSTM, strong on short multivariate windows, cheap on a laptop GPU.
Missing-data handling	Impute only; GRU-D decay	Imputation plus mask and time-since-last features: model-agnostic, so every baseline gets the same information.
Imputer	Linear interpolation, KNN, matrix factorisation	LOCF (as in the baseline) with mean and CART as sensitivity checks; interpolation uses future values and would leak.
Occupancy input	Wi-Fi or camera counts	Plug-load and lighting power: available for the whole period and a direct heat-gain measure; avoids person-related data.
Uncertainty	Full BNN or ensembles for all models	Point forecasts with five seeds; the BNN keeps its probabilistic form to stay faithful to the baseline.

5 Implementation

The artefact is a Python 3.13 code base of five modules and one notebook (Table 5). It uses pandas and NumPy for data handling, scikit-learn (Pedregosa et al., 2011) for the CART imputer, XGBoost 3.2 (Chen and Guestrin, 2016) and PyTorch 2.11 (Paszke et al., 2019). Versions are pinned in `requirements.txt`.

Table 5: Code modules.

File	Role
<code>src/prepare_data.py</code>	Builds the hourly table from the raw files, flags impossible readings and natural gaps, writes the data dictionary and profile.
<code>src/masking.py</code>	MCAR and block masks, LOCF/mean/CART imputers fitted on training rows, mask and time-since-last features.
<code>src/models.py</code>	GRU/LSTM forecasters with optional temperature head, RC physics module, Bayesian layers and BNN, early-stopped training, device selection.
<code>src/experiment.py</code>	Split, tuning, five-seed training and refit, evaluation grid, plausibility test, imputation sensitivity.
<code>src/analysis.py</code>	Bootstrap and Wilcoxon tests, Holm adjustment, all tables and figures.
<code>tests/test_pipeline.py</code>	19 unit tests (leakage, masking, scaling, physics, metrics, split).
<code>notebooks/physics_informed_hvac.ipynb</code>	End-to-end run on Colab, Jupyter, macOS or Windows.

Leakage control. Imputation is strictly causal: LOCF only looks back, and the CART imputer uses other channels at the same hour. A unit test perturbs all values after an hour h and checks that no feature at or before h changes. Scaling and imputation statistics are computed on training rows, and a test checks that changing the test data leaves them unchanged. The split is chronological and a test checks that train, validation and test do not overlap and that every target is observed.

Masks. Masks are generated on the whole timeline from a seed that encodes the purpose (training pool, validation or test), the seed number and the scenario index, so masks are reproducible and identical across models. Tests check that the removed share matches the nominal rate, that block outages are contiguous and at least six hours long, that natural gaps never reappear and that the meter is untouched unless requested.

Training. Recurrent models are trained with AdamW, batch size 256, 120 steps per epoch sampled from the nine-version training pool, gradient clipping at 1.0 and early stopping (patience 6) on the mean validation RMSE over five validation conditions. The BNN uses the baseline’s batch sizes and learning rates. After selection, every model is refitted on train plus validation for the best epoch count (XGBoost: best number of trees).

Hardware and portability. The device is chosen automatically: CUDA, then Apple-silicon MPS, then CPU. The experiments in this report ran on an Apple-silicon laptop with MPS for the recurrent models. XGBoost and PyTorch each ship their own OpenMP runtime; on macOS, loading both in one process crashed or deadlocked PyTorch. The runner therefore trains the tree family and the neural family in separate processes. Because masks depend only on seeds, the families still see identical inputs. The notebook detects whether it is inside the repository, clones only this project with a sparse checkout when it is not (for example on Colab), installs missing packages, and runs every script in a fresh interpreter, which works the same way on Windows, macOS and Linux. A 0.8 MB compressed copy of the hourly table is bundled, so no Kaggle credentials are needed to reproduce the results.

6 Evaluation

All numbers are test-set results (1 July–31 December 2020, 4,414 hours) as mean $\pm$ standard deviation over five seeds, unless stated otherwise. Tuning trials are listed in the **tuning** tables of the results folder; the selected configurations are XGBoost with depth 8 and learning rate 0.05, the BNN with learning rate $3 \cdot 10^{-4}$ and batch 128, the LSTM with 64 units and learning rate $3 \cdot 10^{-3}$, and the GRU backbone with 128 units, learning rate 10^{-3} and dropout 0.1.

6.1 Comparison with the baseline paper

Table 6 and Figure 4 compare every model on complete data with the numbers published by Mahajan et al. (2024) for the same building, target and test window.

Table 6: Complete-data test accuracy (Jul-Dec 2020, hourly HVAC kWh). Published rows are from Mahajan et al. (2024); our rows are mean $\pm$ s.d. over five seeds.

model	MAE	RMSE	MAPE	R2
QRF (published)	17.17	21.67	0.40	–
MC-LSTM (published)	8.21	11.86	0.45	–
BNN (published)	7.06	9.65	0.35	–
XGBoost	6.03 ± 0.29	7.97 ± 0.45	0.291 ± 0.024	0.634 ± 0.042
GRU-aux	5.73 ± 0.64	8.91 ± 0.57	0.215 ± 0.033	0.542 ± 0.059
PI-GRU	5.70 ± 0.43	9.01 ± 0.72	0.216 ± 0.022	0.530 ± 0.076
GRU	5.65 ± 0.68	9.07 ± 0.67	0.205 ± 0.027	0.525 ± 0.069
PI-GRU-nomask	6.01 ± 0.51	9.24 ± 0.65	0.229 ± 0.033	0.507 ± 0.070
LSTM	6.04 ± 0.95	9.44 ± 1.09	0.237 ± 0.041	0.482 ± 0.120
BNN	6.27 ± 0.50	10.42 ± 0.83	0.231 ± 0.017	0.373 ± 0.098
BNN-paper	9.51 ± 0.87	12.11 ± 0.57	0.405 ± 0.012	0.155 ± 0.079
Profile	10.65 ± 0.00	14.28 ± 0.00	0.577 ± 0.000	-0.173 ± 0.000

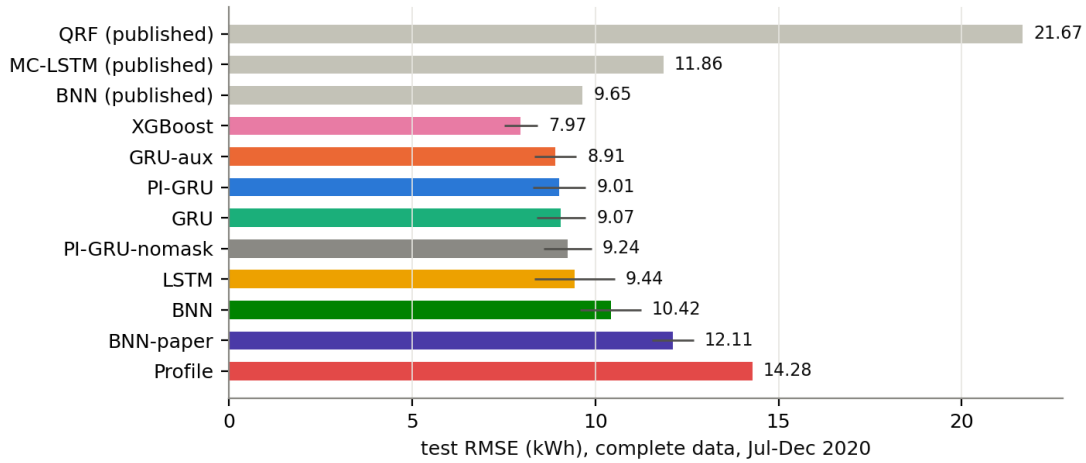


Figure 4: Complete-data test RMSE. Grey: published by Mahajan et al. (2024). Error bars: one standard deviation over five seeds.

The baseline is beaten on every metric. The PI-GRU reaches an RMSE of 9.01 ± 0.72 kWh, an MAE of 5.70 ± 0.43 kWh and a MAPE of 0.216 ± 0.022 , against 9.65, 7.06 and 0.35 for the published BNN: 7% lower RMSE, 19% lower MAE and 38% lower MAPE. XGBoost on the same features has the lowest RMSE (7.97 ± 0.45 , 17% below the published BNN), while the recurrent models have the lowest MAE and MAPE (GRU 5.65 and 0.205). All five recurrent models and XGBoost beat all three published models on all three metrics; only the BNN on our features falls short on RMSE (10.42).

The re-implementation is credible. The BNN re-implemented with the baseline’s own inputs (BNN-paper) scores 12.11 ± 0.57 kWh RMSE, 9.51 MAE and 0.405 MAPE. That lies between the published BNN and QRF and next to the published MC-LSTM (11.86, 8.21, 0.45). The residual gap to the published BNN is plausibly due to the two inputs that are not in the public release (wind speed and the maintenance-event flag) and to training for early-stopped epochs instead of 2,000. Comparing the rows in Table 6 separates the sources of the gain: moving the same BNN from the baseline’s inputs to this project’s inputs (mask, time-since-last and previous-hour features) lowers RMSE from 12.11 to 10.42 (14%); adding 24-hour temporal context with a recurrent network lowers it further to about 9.0 (another 13%).

6.2 Robustness to sensor loss

Table 7 and Figure 5 show how RMSE changes as readings are removed.

Table 7: Test RMSE (kWh) under controlled sensor loss, estimate task, mean of five seeds. Slope: RMSE increase per 10 percentage points of missing readings (mean $\pm$ s.d. over seeds).

model	complete	mcAR 20%	mcAR 40%	block 20%	block 40%	mcAR slope	block slope
BNN	10.42 $\pm$ 0.83	10.05	9.72	10.81	11.39	-0.170 $\pm$ 0.087	0.298 $\pm$ 0.205
BNN-paper	12.11 $\pm$ 0.57	12.16	12.30	12.77	12.97	0.048 $\pm$ 0.013	0.201 $\pm$ 0.123
Profile	14.28 $\pm$ 0.00	14.28	14.28	14.28	14.28	-0.000 $\pm$ 0.000	-0.000 $\pm$ 0.000
XGBoost	7.97 $\pm$ 0.45	8.09	8.25	8.70	9.52	0.070 $\pm$ 0.026	0.373 $\pm$ 0.051
GRU-aux	8.91 $\pm$ 0.57	8.89	8.89	9.59	10.38	-0.003 $\pm$ 0.061	0.417 $\pm$ 0.152
PI-GRU	9.01 $\pm$ 0.72	8.98	8.96	9.78	10.36	-0.010 $\pm$ 0.041	0.418 $\pm$ 0.116
GRU	9.07 $\pm$ 0.67	9.04	9.01	9.77	10.29	-0.007 $\pm$ 0.063	0.373 $\pm$ 0.193
PI-GRU-nomask	9.24 $\pm$ 0.65	9.27	9.40	10.30	11.41	0.035 $\pm$ 0.024	0.543 $\pm$ 0.112
LSTM	9.44 $\pm$ 1.09	9.38	9.30	10.12	10.61	-0.031 $\pm$ 0.051	0.347 $\pm$ 0.234

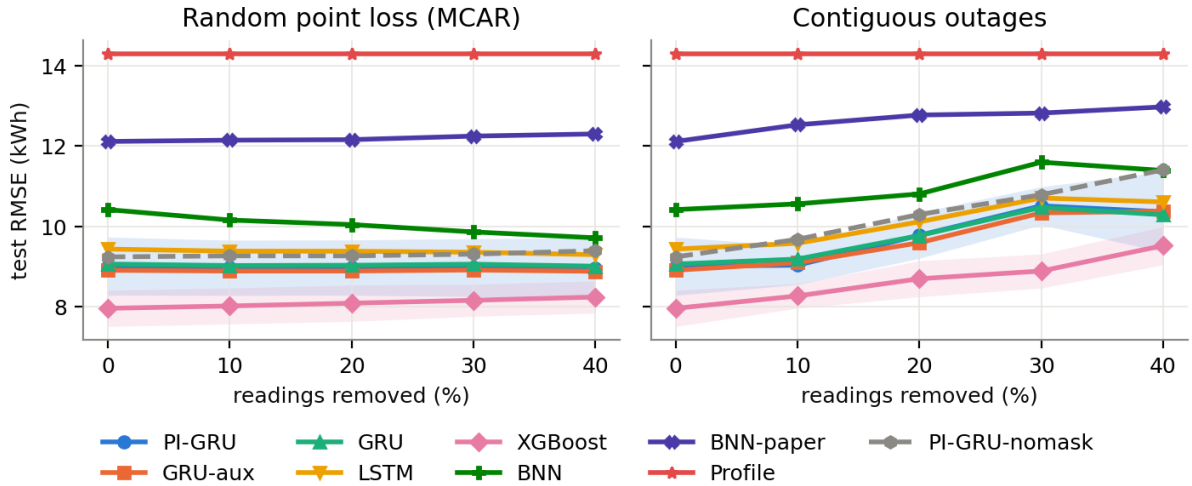


Figure 5: Test RMSE against the share of sensor readings removed (estimation task). Bands: $\pm$ one standard deviation for PI-GRU and XGBoost.

RQ1 and RQ2: random loss is harmless, outages are not. Under MCAR, RMSE is essentially flat for every model up to 40% loss (slopes between -0.03 and $+0.07$ kWh per 10 points). Hourly building signals change slowly, so the last observation carried forward is almost as good as the missing reading, and the mask features tell the model how stale it is. The BNN on our features even improves slightly, which suggests that the masked training pool regularises it. Contiguous outages are different: at 40% loss the RMSE of XGBoost rises by 19% (7.97 to 9.52), of the PI-GRU by 15% (9.01 to 10.36), of GRU-aux by 17% and of the PI-GRU without mask features by 23% (9.24 to 11.41). The baseline model under the same outages (BNN-paper) degrades by only 7%, but from a much worse starting point; at 40% block loss the PI-GRU is still 20% better than it (10.36 against 12.97). Figure 6 shows the relative degradation.

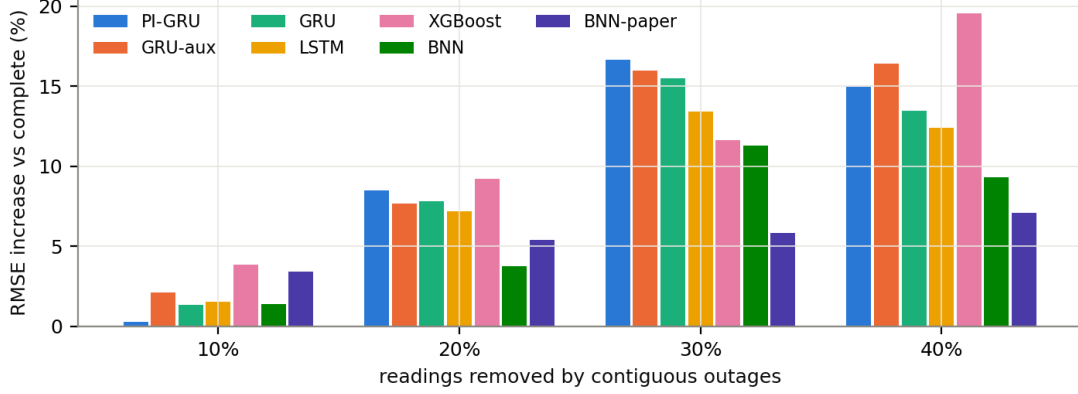


Figure 6: RMSE increase over complete data under contiguous outages.

6.3 Is the gain due to physics? Statistical tests and ablations

Table 8 tests the PI-GRU against the best conventional model at each condition (always XGBoost) and against its architecture-matched twin without the physics term (GRU-aux).

Table 8: PI-GRU minus reference, estimate task. Negative values favour PI-GRU. CI: day-block bootstrap (2,000 resamples) on seed-averaged errors; p : Wilcoxon signed-rank on daily MSE, Holm-adjusted.

condition	reference	Δ RMSE	$\Delta\%$	95% CI	p_{Holm}
none 0%	XGBoost	+1.06	+13.3	[-0.23, +2.33]	0.429
none 0%	GRU-aux	+0.11	+1.2	[-0.04, +0.26]	1
mcar 10%	XGBoost	+0.96	+11.9	[-0.26, +2.39]	0.49
mcar 10%	GRU-aux	+0.09	+1.0	[-0.07, +0.23]	1
mcar 20%	XGBoost	+0.90	+11.1	[-0.30, +2.25]	0.572
mcar 20%	GRU-aux	+0.09	+1.0	[-0.06, +0.24]	1
mcar 30%	XGBoost	+0.83	+10.1	[-0.39, +2.18]	0.907
mcar 30%	GRU-aux	+0.07	+0.8	[-0.08, +0.21]	1
mcar 40%	XGBoost	+0.73	+8.9	[-0.48, +2.10]	1
mcar 40%	GRU-aux	+0.08	+0.9	[-0.07, +0.23]	1
block 10%	XGBoost	+0.77	+9.3	[-0.27, +1.89]	1
block 10%	GRU-aux	-0.06	-0.6	[-0.20, +0.09]	1
block 20%	XGBoost	+1.08	+12.4	[-0.05, +2.29]	1
block 20%	GRU-aux	+0.19	+2.0	[+0.01, +0.38]	1
block 30%	XGBoost	+1.62	+18.2	[+0.54, +2.84]	1
block 30%	GRU-aux	+0.18	+1.7	[-0.01, +0.37]	1
block 40%	XGBoost	+0.87	+9.1	[-0.05, +1.81]	0.22
block 40%	GRU-aux	-0.00	-0.0	[-0.15, +0.13]	1

H1 is rejected. XGBoost has a lower RMSE than the PI-GRU at every condition, by 9–18%. The day-block bootstrap intervals include zero at most conditions and no difference survives the Holm adjustment, but XGBoost is better in 40 of the 45 seed-condition pairs. The pre-registered success criterion (5% lower RMSE than the best conventional model at 20–40% loss) is not met.

H2 is not supported. The PI-GRU and GRU-aux differ by between -0.6% and $+2.0\%$ in RMSE, with intervals that straddle zero and Holm-adjusted p -values of 1.0. With the validation-selected weight $\lambda = 0.03$, the physics term neither helps nor hurts accuracy. Larger weights hurt (Figure 7): validation RMSE rises monotonically from 5.36 at $\lambda = 0.03$ to 7.42 at $\lambda = 3$, against 5.53 for $\lambda = 0$.

Mask awareness matters more than physics. Removing the mask and time-since-last

features (PI-GRU-nomask) raises the RMSE under 40% block outages from 10.36 to 11.41 (10%) and the outage slope from 0.42 to 0.54 kWh per 10 points. This is the clearest single ablation effect in the study.

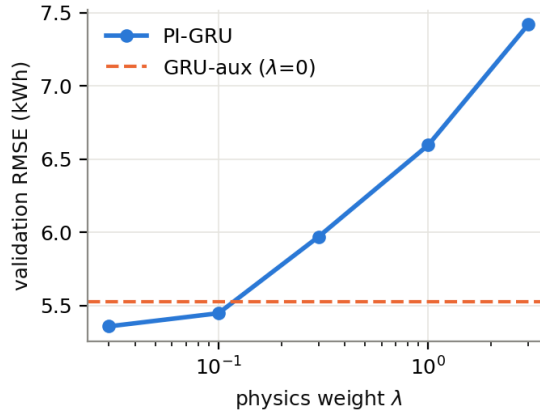


Figure 7: Validation RMSE against the physics weight λ (seed 0).

6.4 Physical consistency and learned parameters

Table 9: Physical consistency, estimate task. Response of predicted HVAC energy to +1 K outdoor temperature in cooling hours, share of cooling hours where the response is negative, and RMS residual (K/h) of the least-squares RC balance for dual-head models.

model	$\partial E / \partial T_{out}$ (kWh/K)	implausible (%)	RC residual block 40%	RC residual mcar 40%	RC residual none 0%
BNN	0.61 ± 0.28	18.8 ± 18.5	–	–	–
BNN-paper	1.26 ± 0.19	7.5 ± 6.6	–	–	–
GRU	0.59 ± 0.12	33.3 ± 5.8	–	–	–
GRU-aux	0.70 ± 0.06	28.2 ± 10.0	0.268	0.130	0.124
LSTM	0.62 ± 0.08	15.3 ± 4.8	–	–	–
PI-GRU	0.73 ± 0.13	29.4 ± 4.5	0.266	0.126	0.112
PI-GRU-nomask	0.54 ± 0.18	35.0 ± 10.6	0.315	0.142	0.138
XGBoost	0.33 ± 0.02	7.4 ± 1.7	–	–	–

H3 is partly supported. On complete data the physics term lowers the RC-balance residual of the PI-GRU by 10% relative to GRU-aux (0.112 against 0.124 K/h) at a cost of 1.1% in RMSE, inside the 2% tolerance. The advantage shrinks to 3% at 40% MCAR loss and disappears under 40% block outages (0.266 against 0.268). Without mask features the residual is highest (0.138 and 0.315). The outdoor-temperature test tells a less favourable story: every model predicts, on average, more HVAC energy when the outdoor temperature rises (0.33–1.26 kWh/K in cooling hours), which is physically right, but the recurrent models do the opposite in 28–35% of individual cooling hours, against 7% for XGBoost and the baseline BNN. The physics term does not reduce this share (29.4% against 28.2%).

Table 10: Learned RC coefficients (estimate task, mean $\pm$ s.d. over five seeds) against a least-squares fit on clean training data.

model	τ (h)	b	c	d	e
PI-GRU	48.9 ± 6.2	0.0025 ± 0.0005	0.0055 ± 0.0008	0.0045 ± 0.0005	0.1957 ± 0.0029
PI-GRU-nomask	39.1 ± 9.4	0.0016 ± 0.0007	0.0074 ± 0.0016	0.0060 ± 0.0012	0.1962 ± 0.0037
Least-squares RC (train)	n.i. ($a=-1.6e-05$)	0.0005	0.0027	0.1399	-0.0527

The learned time constant of the PI-GRU is $\tau = 48.9 \pm 6.2$ h, which is within the range expected

for a steel-framed office building with a curtain wall, and it is stable across seeds. However, a least-squares fit of the same RC balance on clean training data gives a conductance of essentially zero (and of the wrong sign), so the time constant is *not identifiable* from building-level data. The indoor temperature is held within a narrow band by the HVAC system (standard deviation 0.8 K over the whole period), and the rooftop-unit meter covers only part of the heating and cooling plant. The learned τ therefore reflects the positivity constraint and the initialisation as much as the data (Figure 8). This is the main reason why the physics term cannot add information here: the prior is too weakly coupled to the observed energy.

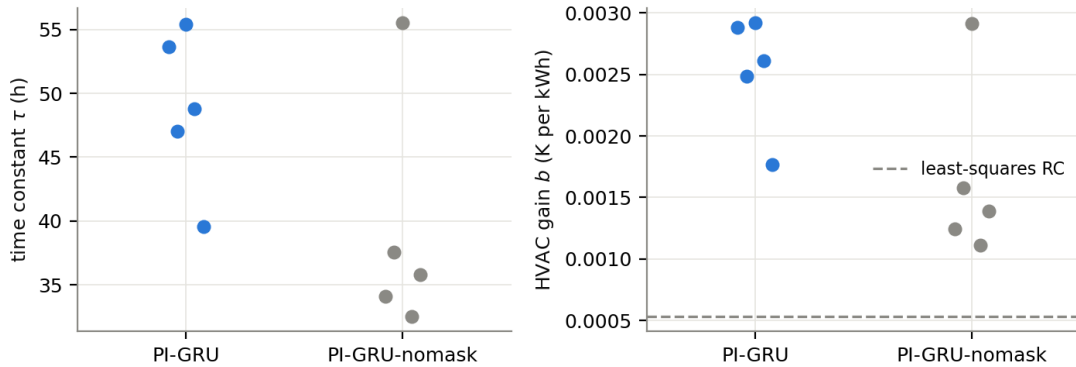


Figure 8: Learned time constant and HVAC gain across five seeds.

6.5 Operating strata

Table 11: Test RMSE (kWh) by operating stratum, estimate task, mean of five seeds.

model	condition	occupied	unoccupied	mild T_out	extreme T_out
PI-GRU	none 0%	9.160000	8.930000	7.110000	14.260000
PI-GRU	block 40%	10.830000	10.090000	9.160000	14.160000
GRU-aux	none 0%	9.000000	8.860000	7.120000	13.930000
GRU-aux	block 40%	10.870000	10.080000	9.180000	14.150000
XGBoost	none 0%	8.250000	7.800000	7.290000	10.220000
XGBoost	block 40%	10.100000	9.180000	9.110000	11.020000
BNN-paper	none 0%	11.860000	12.250000	10.950000	15.890000
BNN-paper	block 40%	13.090000	12.890000	12.000000	16.190000

The strata explain why XGBoost wins on RMSE while the recurrent models win on MAE. In mild outdoor conditions (between the 10th and 90th percentile of the test period) the PI-GRU is slightly better than XGBoost on complete data (7.11 against 7.29 kWh). In extreme conditions its error is 40% higher (14.26 against 10.22): the recurrent models extrapolate poorly to temperatures outside the range seen in training, and a few large errors dominate the RMSE. Occupied and unoccupied hours behave alike. Figure 9 shows an August week.

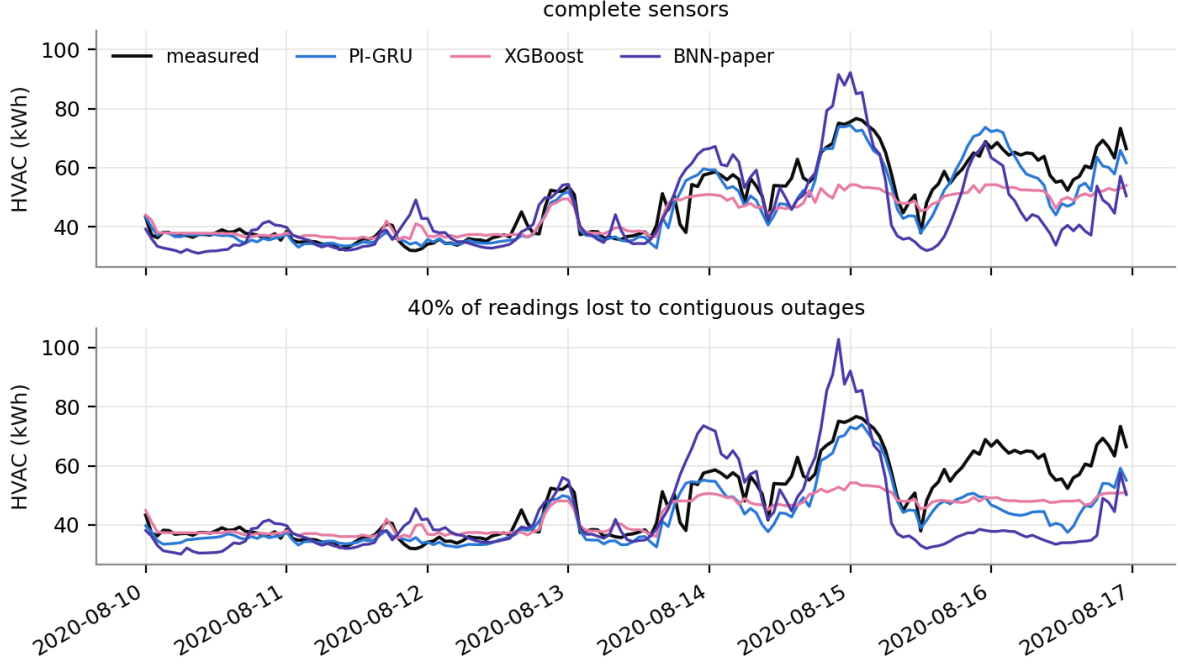


Figure 9: Measured and predicted HVAC energy for one test week (seed 0), with complete sensors and with 40% of readings lost to contiguous outages.

6.6 Choice of imputer

Table 12: Test RMSE (kWh) by imputation method, estimation task, mean of three seeds.

model	imputer	block 20%	block 40%	mcar 20%	mcar 40%	none 0%
GRU-aux	cart	9.280000	10.440000	8.920000	8.350000	9.250000
GRU-aux	locf	9.590000	10.160000	8.630000	8.620000	8.610000
GRU-aux	mean	9.730000	11.930000	9.290000	9.210000	9.260000
PI-GRU	cart	9.720000	10.680000	9.370000	8.550000	9.860000
PI-GRU	locf	9.830000	10.110000	8.810000	8.750000	8.820000
PI-GRU	mean	9.460000	11.290000	8.830000	8.720000	8.810000
XGBoost	cart	8.310000	9.040000	7.930000	8.450000	7.560000
XGBoost	locf	8.780000	9.490000	8.080000	8.190000	7.960000
XGBoost	mean	9.210000	10.830000	8.660000	9.700000	8.050000

Table 12 repeats the key conditions with mean and iterative CART imputation (three seeds). Mean imputation is clearly the worst under contiguous outages: at 40% block loss it raises the RMSE of XGBoost to 10.83, of GRU-aux to 11.93 and of the PI-GRU to 11.29, against 9.49, 10.16 and 10.11 with LOCF, because a long outage then shows the model a constant, unrealistic value. CART, which predicts a missing channel from the other channels at the same hour, helps XGBoost at every condition (9.04 at 40% block loss) but not the recurrent models, which already see the neighbouring hours and do best with LOCF under heavy outages. The ranking of the models does not change with the imputer, and the physics term does not make the PI-GRU less sensitive to the imputer than GRU-aux. Imputation choice matters as much as model choice under outages, in line with Lee et al. (2024).

6.7 Computational cost

Table 13: Computational cost, estimate task (mean over seeds).

model	train (s)	epochs / trees	inference (μ s/sample)
BNN	33.3 ± 19.9	11	44.3
BNN-paper	13.3 ± 2.3	3	43.0
GRU	38.6 ± 20.2	10	9.1
GRU-aux	45.4 ± 15.5	12	5.1
LSTM	6.6 ± 1.8	4	12.0
PI-GRU	50.7 ± 10.7	13	4.6
PI-GRU-nomask	34.4 ± 18.2	8	4.5
XGBoost	0.9 ± 0.1	20	0.2

The physics term adds no inference cost, because the residual is only evaluated during training, and about 12% to training time relative to GRU-aux. All recurrent models train in under a minute per seed on the laptop GPU and predict in under 10μ s per hour. XGBoost is two orders of magnitude cheaper to train and predict; the Bayesian networks are the slowest at inference because they average 30 posterior samples.

6.8 Supplementary: forecasting with meter history

When the previous hours of HVAC energy are available as input, the problem changes character (Table 14, Figure 10). Persistence alone reaches an RMSE of 3.75 kWh, less than half of the best estimation error, and only XGBoost matches it (3.74); the recurrent models are 3–9% worse (3.85–4.07), and the PI-GRU (3.95) is again level with GRU-aux (3.92). Losing up to 40% of the other sensors changes almost nothing, because the meter history carries most of the information. Losing the meter itself is different: with 40% of all channels, including the meter, removed by outages, every model’s RMSE roughly doubles (7.05–7.32 for the recurrent models and XGBoost against 7.56 for persistence). The recurrent models and XGBoost then beat persistence by 3–7%, because they can fall back on the building sensors; the Bayesian networks do not. The physics weight selected on validation for this task was $\lambda = 0.1$. Here the physics term costs a little accuracy: the PI-GRU is 1–2% worse than GRU-aux under sensor loss, significant after Holm adjustment at several conditions (`significance_forecast` table in the results folder), and about 1% better under meter outages, which is not significant.

Table 14: Test RMSE (kWh) under controlled sensor loss, forecast task, mean of five seeds. Slope: RMSE increase per 10 percentage points of missing readings (mean $\pm$ s.d. over seeds).

model	complete	mcar 20%	mcar 40%	block 20%	block 40%	mcar slope	block slope
BNN-paper	12.10 ± 0.55	12.15	12.28	12.78	12.96	0.047 ± 0.011	0.199 ± 0.125
XGBoost	3.74 ± 0.02	3.75	3.76	3.77	3.78	0.004 ± 0.001	0.010 ± 0.005
Persistence	3.75 ± 0.00	3.75	3.75	3.75	3.75	-0.000 ± 0.000	-0.000 ± 0.000
GRU	3.85 ± 0.06	3.90	3.93	3.92	3.91	0.019 ± 0.009	0.020 ± 0.010
GRU-aux	3.92 ± 0.05	3.97	4.02	3.99	4.09	0.025 ± 0.009	0.042 ± 0.007
PI-GRU	3.95 ± 0.10	4.02	4.08	4.08	4.14	0.032 ± 0.014	0.048 ± 0.019
PI-GRU-nomask	3.99 ± 0.05	4.02	4.02	4.04	4.07	0.008 ± 0.004	0.019 ± 0.011
LSTM	4.07 ± 0.20	4.10	4.13	4.10	4.09	0.018 ± 0.012	0.015 ± 0.021
BNN	4.65 ± 0.36	4.75	4.83	4.83	4.83	0.042 ± 0.025	0.042 ± 0.069
Seasonal	9.97 ± 0.00	9.97	9.97	9.97	9.97	-0.000 ± 0.000	-0.000 ± 0.000

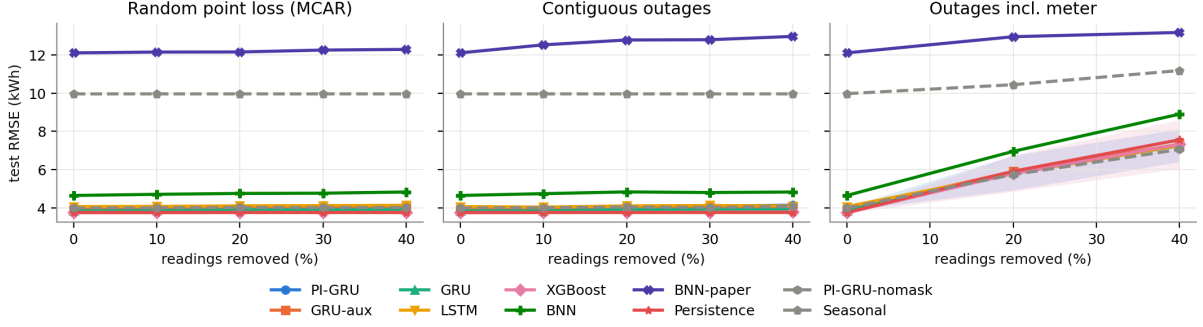


Figure 10: Forecasting task: test RMSE against the share of readings removed, including outages that also hit the HVAC meter.

6.9 Summary against the objectives and hypotheses

Table 15: Objectives and hypotheses against the evidence.

Item	Outcome	Evidence
O1 Reproducible benchmark	Achieved	Seeded masks shared by all models; 19 unit tests; one-command pipeline and notebook
O2 Baseline and models	Achieved	BNN re-implemented within the range of published results; 8 models and 2 ablations
O3 Rigorous evaluation	Achieved	5 seeds, day-block bootstrap, Holm-adjusted Wilcoxon tests, strata, physics checks, cost
O4 Beat the published baseline	Achieved	PI-GRU: RMSE -7% , MAE -19% , MAPE -38% ; XGBoost: RMSE -17%
H1 PI-GRU beats best conventional at 20–40% loss	Rejected	XGBoost 9–18% lower RMSE at every condition
H2 Physics term improves accuracy	Not supported	PI-GRU vs GRU-aux within $\pm 2\%$, all $p_{Holm} = 1$
H3 Lower residual, $\leq 2\%$ accuracy cost	Partly supported	Residual -10% at complete data, $+1.1\%$ RMSE; no residual gain under heavy outages

7 Discussion

7.1 Answer to the research question

The mask-aware PI-GRU improves on the published baseline for Building 59 by a clear margin: 7% lower RMSE, 19% lower MAE and 38% lower MAPE on complete data, and 20% lower RMSE than the re-implemented baseline when 40% of the readings are lost to contiguous outages. The improvement, however, does not come from the physics term. The architecture-matched twin without physics is as accurate, and a well-tuned gradient-boosted tree on the same features is more accurate still in RMSE. What carries the gain is (i) the richer, mask-aware input representation, which lowers the baseline network’s error by 14% on its own, (ii) temporal context, which adds another 13%, and (iii) mask awareness under outages, which is worth 10% at 40% block loss. The answer to the research question is therefore: *physics-informed learning, as implemented here, improves the prediction only as part of a mask-aware temporal model; the physics term itself is neutral for accuracy in the estimation task, costs 1–2% in the forecasting task, and gives a modest reduction in heat-balance residual on complete data.*

RQ1. Random point loss up to 40% has almost no effect on any model, because hourly building signals are smooth and the imputed value is close to the true one. **RQ2.** Contiguous multi-

sensor outages are the real threat, raising RMSE by 15–23% at 40%; this confirms the argument of Lee et al. (2024) that the missingness mechanism, not only the rate, decides the impact, and it means that robustness studies that only use MCAR masks overstate robustness. **RQ3.** The physics term lowers the RC residual by 10% on complete data, but not under heavy outages, and it does not reduce implausible responses to outdoor temperature; the learned time constant is stable ($\tau \approx 49$ h) but not identifiable from the data.

7.2 Why the physics prior did not help here

The literature reports that physics priors help most when data are scarce or when models must extrapolate (Gokhale et al., 2022; Liguori et al., 2024). Three features of this case weaken the prior. First, the indoor temperature is tightly controlled, so the temperature change carries little information about HVAC energy; a least-squares fit cannot even identify the envelope conductance. Second, the HVAC meter covers the rooftop units only, while fan-coil units and the central chilled- and hot-water plant also condition the space, so the balance of Eq. (1) links the temperature to only part of the thermal input. Third, there are 2.5 years of hourly training data, and the network already learns the dominant relationships from data. This matches the open problems named by Ma et al. (2025): the choice of physics prior and its match to what is metered decide whether PIML helps, and a mis-matched prior can only be down-weighted. The monotonic loss of accuracy as λ grows is direct evidence of that mismatch. A physics-derived feature layer (fan affinity law and air-side sensible load) was also tried in a pilot with XGBoost and gave no consistent gain across seeds, so it was not adopted.

7.3 Comparison with the literature

The complete-data results agree with Mahajan et al. (2024) that their BNN is more accurate than an LSTM trained on the same per-hour inputs; they add that a recurrent network with a 24-hour window and mask features beats both. They agree with Che et al. (2018) that exposing the missingness pattern to the model is valuable. They partly agree with Liguori et al. (2024): the physics term has little effect on accuracy with complete data, but unlike in their imputation study it does not buy graceful degradation for energy prediction. The strongest model, XGBoost, is consistent with the strong record of gradient-boosted trees on tabular prediction problems reported by Chen and Guestrin (2016).

7.4 Implications

For practitioners. (1) Test energy models with contiguous multi-sensor outages, not just random gaps. (2) Give models the observation mask and the time since the last reading; it is cheap and it is the most effective single measure found here. (3) A gradient-boosted model with these features is a strong, fast default (under a second to train, $0.2 \mu\text{s}$ per prediction); a recurrent model is preferable when typical-hour accuracy (MAE, MAPE) matters more than rare large errors. (4) Before adding a physics term, check that the metered energy and the modelled heat balance cover the same equipment.

For researchers. Physics-informed energy models should be reported with an architecture-matched $\lambda = 0$ ablation; without it, the gain of the PI-GRU over the baseline would have been wrongly credited to physics. Identifiability of the physical parameters should be checked with a plain least-squares fit, because a stable learned parameter can be a product of the constraint rather than the data.

7.5 Limitations

- One building, one mild climate and one test period, which is also affected by COVID-19 occupancy changes; the conclusions about the physics prior may not transfer to buildings with a metered plant or larger temperature swings.
- The published baseline numbers come from the paper; two of its inputs are not in the public release, so the re-implementation is close but not exact.
- The RC model is first order and uses a mode proxy derived from rooftop-unit air temperatures; zone-level flows, chilled-water and hot-water energy are not available.
- Masks are synthetic. They are designed to resemble real outages (contiguous, multi-channel), but real failure modes such as drift or stuck values are not simulated.
- Five seeds give wide intervals for small effects; differences of about 1% between the PI-GRU and its twin cannot be resolved.
- Hyperparameter budgets are equal but small (five trials per family); larger searches could change the ranking of the recurrent models.

8 Conclusion and Future Work

This project asked whether a physics-informed, mask-aware recurrent network predicts hourly HVAC energy more accurately than conventional models, including the published baseline, when a building’s sensors are incomplete. On LBNL Building 59, with the baseline’s own target, split and metrics, the PI-GRU beats the published BNN of Mahajan et al. (2024) on every metric (RMSE 9.01 against 9.65 kWh, MAE 5.70 against 7.06 kWh, MAPE 0.22 against 0.35) and stays 20% better than the re-implemented baseline under 40% contiguous sensor outages. XGBoost on the same mask-aware features is more accurate still in RMSE (7.97 kWh).

The objectives were met. A reproducible benchmark with identical seeded masks was built (O1); the baseline was re-implemented and compared with five conventional models and two ablations (O2); accuracy, robustness, physical consistency and cost were evaluated with five seeds, bootstrap intervals and Holm-adjusted tests (O3); and the baseline was beaten on its own protocol (O4). The hypotheses were answered honestly: the PI-GRU did not beat the best conventional model (H1 rejected), the physics term did not improve accuracy over its matched twin (H2 not supported), and it lowered the heat-balance residual on complete data without a meaningful accuracy cost (H3 partly supported). Random sensor loss proved almost harmless and contiguous outages harmful, and mask awareness, not physics, gave the largest robustness gain.

Future work.

- Replicate on buildings where the metered energy covers the whole plant and where indoor temperatures vary more, so that the RC parameters are identifiable.
- Use a second-order RC model with zone temperatures, or a hard-constrained architecture such as a physically consistent network (Di Natale et al., 2022), where zone-level data allow it.
- Add realistic sensor faults (drift, stuck values, bias) to the outage generator.
- Combine the strong tabular model with the recurrent model, for example by stacking, and add calibrated uncertainty under sensor loss, which was the focus of the baseline.
- Evaluate transfer across buildings from the same data programme.

References

- Kadir Amasyali and Nora M. El-Gohary. A review of data-driven building energy consumption prediction studies. *Renewable and Sustainable Energy Reviews*, 81:1192–1205, 2018. doi: 10.1016/j.rser.2017.04.095.
- Peder Bacher and Henrik Madsen. Identifying suitable models for the heat dynamics of buildings. *Energy and Buildings*, 43(7):1511–1522, 2011. doi: 10.1016/j.enbuild.2011.02.005.
- Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight Uncertainty in Neural Networks. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1613–1622, 2015.
- Zhengping Che, Sanjay Purushotham, Kyunghyun Cho, David Sontag, and Yan Liu. Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8(1):6085, 2018. doi: 10.1038/s41598-018-24271-9.
- Tianqi Chen and Carlos Guestrin. Xgboost. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.
- Yongbao Chen, Qiguo Yang, Zhe Chen, Chengchu Yan, Shu Zeng, and Mingkun Dai. Physics-informed neural networks for building thermal modeling and demand response control. *Building and Environment*, 234:110149, 2023. doi: 10.1016/j.buildenv.2023.110149.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734, 2014. doi: 10.3115/v1/D14-1179.
- L. Di Natale, B. Svetozarevic, P. Heer, and C.N. Jones. Physically consistent neural networks for building thermal modeling: Theory and analysis. *Applied Energy*, 325:119806, 2022. doi: 10.1016/j.apenergy.2022.119806.
- Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–1059, 2016.
- Gargya Gokhale, Bert Claessens, and Chris Develder. Physics informed neural networks for control oriented thermal modeling of buildings. *Applied Energy*, 314:118852, 2022. doi: 10.1016/j.apenergy.2022.118852.
- Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.
- Sture Holm. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- Tianzhen Hong, Na Luo, David Blum, and Zhe Wang. A three-year building operational performance dataset for informing energy efficiency. Dryad, dataset, 2022.
- George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3(6):422–440, 2021. doi: 10.1038/s42254-021-00314-5.

- Jiwon Kim, Younghoon Kwak, Sun-Hye Mun, and Jung-Ho Huh. Imputation of missing values in residential building monitored data: Energy consumption, behavior, and environment information. *Building and Environment*, 245:110919, 2023. doi: 10.1016/j.buildenv.2023.110919.
- Hans R. Kunsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1989. doi: 10.1214/aos/1176347265.
- Kyungjae Lee, Hyunwoo Lim, Jeongyun Hwang, and Doyeon Lee. Evaluating missing data handling methods for developing building energy benchmarking models. *Energy*, 308:132979, 2024. doi: 10.1016/j.energy.2024.132979.
- Antonio Liguori, Matias Quintana, Chun Fu, Clayton Miller, Jérôme Frisch, and Christoph van Treeck. Opening the black box: Towards inherently interpretable energy data imputation models using building physics insight. *Energy and Buildings*, 310:114071, 2024. doi: 10.1016/j.enbuild.2024.114071.
- Na Luo, Zhe Wang, David Blum, Christopher Weyandt, Norman Bourassa, Mary Ann Piette, and Tianzhen Hong. A three-year dataset supporting research on building energy management and occupancy analytics. *Scientific Data*, 9(1):156, 2022. doi: 10.1038/s41597-022-01257-x.
- Zhihao Ma, Gang Jiang, Yuqing Hu, and Jianli Chen. A review of physics-informed machine learning for building energy modeling. *Applied Energy*, 381:125169, 2025. doi: 10.1016/j.apenergy.2024.125169.
- Akash Mahajan, Srijita Das, Wencong Su, and Van-Hai Bui. Bayesian-neural-network-based approach for probabilistic prediction of building-energy demands. *Sustainability*, 16(22):9943, 2024. doi: 10.3390/su16229943.
- Vasilis Michalakopoulos, Sotiris Pelekis, Giorgos Kormpakis, Vagelis Karakolis, Spiros Mouzakis, and Dimitris Askounis. Data-driven building energy efficiency prediction using physics-informed neural networks. In *2024 IEEE Conference on Technologies for Sustainability (SusTech)*, pages 84–91, 2024. doi: 10.1109/SusTech60925.2024.10553513.
- Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. PyTorch: An Imperative Style, High-Performance Deep Learning Library. In *Advances in Neural Information Processing Systems*, volume 32, pages 8024–8035, 2019.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- M. Raissi, P. Perdikaris, and G.E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.
- DONALD B. RUBIN. Inference and missing data. *Biometrika*, 63(3):581–592, 1976. doi: 10.1093/biomet/63.3.581.